

Import: components, validation, and writes

Status: Working. The write path cannot start until local validation succeeds and the user accepts a confirmation dialog.

User and data flow

- **1. Upload:** File input accepts CSV. The file is read with `File.text()`, parsed in memory, and capped at 10,000 data rows. Empty/duplicate headers and unclosed quotes are rejected.
- **2. Target:** Object choices come from global describe and are limited to createable/updateable objects. Insert, Update, and Upsert are explicit operations.
- **3. Mapping:** Headers auto-map by case-insensitive API name or label; each column may be remapped or ignored. Duplicate target fields are rejected.
- **4. Validation:** Conversion checks booleans, numbers, ISO dates, datetimes, text lengths, restricted picklists, writeability, required create fields, Id for Update, and the selected mapped external ID for Upsert.
- **5. Preview:** The first ten converted rows are shown before submission; validation issues include CSV row and field.
- **6. Confirmation:** The modal names the operation, object record count, and connected org/user, and warns that Salesforce rules and automation apply.
- **7. Execution:** Records are processed in client-side batches of 25 with bounded client concurrency. Insert uses `POST sObject`; Update uses `PATCH` by Id; Upsert uses `PATCH` by external-ID field/value.
- **8. Results:** Progress updates after each batch. Every row receives success/id or an error string. A downloadable row-level CSV contains row, success, id, and error.

Salesforce APIs and access enforcement

Purpose	API / endpoint	Enforcement
Discover objects	REST GET <code>/subjects/</code>	Object visibility and createable/updateable flags are those of the connected user.
Discover fields	REST GET <code>/subjects/{object}/describe/</code>	Mapping choices use createable/updateable/externalId and other describe attributes.
Insert	REST POST <code>/subjects/{object}/</code>	Salesforce applies object create access, FLS, requiredness, validation, triggers, Flows, and duplicate rules.
Update	REST PATCH <code>/subjects/{object}/{Id}</code>	Salesforce applies object update, record access/sharing, FLS, locks, validation, and automation.
Upsert	REST PATCH <code>/subjects/{object}/{externalIdField}/{value}</code>	Salesforce resolves the external ID and applies create or update authorization and record access.

Error handling and cancellation

- File and local validation errors block the confirmation button; Salesforce row failures do not stop unrelated rows.
- `AbortController` cancels outstanding fetches. Authentication refresh, typed API errors, network errors, and rate-limit usage tracking remain centralized in `SalesforceClient`.
- Import history is saved only after a completed run and contains aggregate metadata, never source rows or CSV contents.